

GATE C PROTECTED REVIEW - Mobile Onboarding

RC2 | Recomputed controls | Threshold/CI distinction | State machine unchanged

Open only after a genuine saved Gate attempt. This review diagnoses the attempt; it must not be used as a pre-attempt solution.

Layer 1 - Data-integrity controls

Control	Expected
Rows / units	2400
Duplicate unit IDs	0
Missing unit IDs	0
Invalid group labels	0
Invalid converted labels	0
Invalid support-contact labels	0
Control n	1200
Treatment n	1200

Layer 2 - Statistical controls

Metric	Expected
Control conversions	96
Treatment conversions	132
Control rate	8.0000%
Treatment rate	11.0000%
Absolute lift	3.0000pp
Relative lift	37.5000%
z statistic	2.5062
two-sided p-value	0.0122
95% CI for rate difference	[0.6569pp, 5.3431pp]
Support contacts	Control 60 Treatment 105
Support rates	5.0000% vs 8.7500%
Support change	3.7500pp

Layer 3 - Threshold and uncertainty contract

Question	Result
Minimum worthwhile effect	1.50pp
Observed point estimate clears MWE?	True
95% CI lower bound clears MWE?	False
Guardrail maximum increase	1.00pp
Guardrail breached?	True

Important distinction: "point estimate exceeds MWE" and "the uncertainty interval establishes at least the MWE" are different claims. M08 should never collapse them into one boolean.

Expected decision language

DO NOT LAUNCH AS-IS. Primary-metric evidence is positive and the observed +3.0pp clears the 1.5pp point threshold, but the support guardrail is breached by +3.75pp. The CI lower bound also does not establish a full ≥ 1.5 pp effect.

Critical interpretation checks

- State the null hypothesis and alpha separately from the estimated effect.
- Use p-value language conditional on H_0 ; never write that p is the probability H_0 is true.
- Use percentage points for absolute rate differences and percent for relative lift.
- Treat the 95% CI as uncertainty around the effect estimate; do not claim a fixed parameter has a 95% posterior probability unless a Bayesian model was actually used.
- A statistically significant result is not automatically business-important. A business-important point estimate is not automatically statistically established.
- Guardrails can block rollout even when the primary metric improves.

Layer 4 - Diagnostic decision tree

- **Counts/labels wrong:** stop. Repair data integrity before calculating rates.
- **Rates wrong:** check numerator/denominator and whether percentages use the full assigned group.
- **pp vs % wrong:** absolute lift = $\text{Treatment\%} - \text{Control\%}$; relative lift = $\text{absolute lift} / \text{Control\%}$.
- **z/p wrong:** verify pooled SE for the null test and a two-sided p-value.
- **CI wrong:** verify the unpooled SE for the rate difference and keep units in percentage points.
- **Decision too strong:** separately evaluate design/data quality, effect estimate, uncertainty/statistical evidence, MWE and guardrail.
- **Conclusion copied from review:** no pass. Close review and complete changed fresh evidence.

Minimum evidence before pass

Evidence	Pass condition
Saved Gate workbook	Fresh identifiable first attempt exists
Quality block	Counts, duplicate/missing/invalid checks stated
Primary metric	Rates + absolute pp + relative % effect
Uncertainty	95% CI with correct units and interpretation
Hypothesis test	H_0 , alpha, z, p and reject/fail-to-reject language
Business threshold	Observed point estimate compared with MWE; no overclaim about CI
Guardrail	Both rates + pp change + explicit breach/no-breach
Explain-back	60-90 sec own-words defense without false p/CI/causal claims

Targeted repair routing

Failure pattern	Repair focus
Counts, duplicates, invalid assignment/outcome labels	STAT9 experiment quality + data integrity
Rates/denominators or percentage-point units	STAT1 metric discipline + STAT7 effect size
CI arithmetic or probability-style CI wording	STAT4-STAT5 SE / uncertainty
p-value meaning, H0 or reject/fail-to-reject language	STAT6 hypothesis testing
Observed effect confused with guaranteed MWE	STAT7 business threshold + uncertainty
Guardrail ignored because primary metric improved	STAT7 + STAT9 decision contract
Causal claim exceeds randomized design/evidence	STAT8-STAT9 causal limits/design

Before a fresh retest

- Keep the original failed attempt; do not overwrite it.
- Write the exact competency that failed and the repair evidence produced.
- Close this protected review before starting changed evidence.
- Use a changed scenario or assigned Gate; copying the same controls is not mastery.

Fresh retry and state route

Gate C is remediation-only. Clean C pass ends M08. There is no Gate D.

Close this review before the next attempt. Repair only the named weak competency; do not retake by reproducing the same control table.